Subject Code: KCA053

Paper ID : 250367

Roll No: | | | | | | | | | | | | | |

MCA
(SEM IV) THEORY EXAMINATION 2024-25
NATURAL LANGUAGE PROCESSING

TIME: 3 HRS **M.MARKS: 100**

Note: Attempt all Sections. In case of any missing data, choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q No.	Question	CO	Level
a.	List the applications ofNLP.	1	K1
b.	Explain open and closed word classes in English language.	1	K2
c.	Explain knowledge representation.	2	K2
d.	Differentiate between syntax and semantics.	2	K3
e.	Describethe parsing process.	3	K2
f.	Discuss is the function of the lexicon during parsing?	3	K2
g.	Discuss Auxiliary verb with suitable example.	4	K2
h.	Explain different ways to create words from morphemes	4	K2
i.	Explain the use of language model?	5	K2
j.	What is meant by Part-of-Speech (POS) tagging?	5	K1

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

Q No.	Question	CO	Level
a.	Explain the organization of natural language understanding system.	1	K2
b.	Discuss the language generation with respect to machine translation.	2	K2
c.	Describe various syntactic parsing techniques.	3	K2
d.	Determine context free grammar. Discuss its use and limitations.	4	K3
e.	Discuss thedetails about the encoding ambiguity in logical form.	5	K2

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

Q No.	Question	CO	Level
a.	Explain the different steps in natural language understanding in detail.	1	K2
b.	Explain the different levels of language analysis process.	1	K2

4. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Explain top-down parsing and bottom-up parsing with an example.	2	K2
b.	Explain all the steps that are used in machine translation.	2	K2

5. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Discuss morphological analysis in detail.	3	K2
b.	Discuss transition network. Explain the different types of transition network.	3	K2

6. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Illustrate human preferences in parsing in detail.	4	K3
b.	Illustrate lexical analysis. Explain in brief. Also explain finite state automaton.	4	K3

7. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Describe is semantic analysis? Discuss different semantic relationships between the words.	5	K2
b.	Describe generative models and compare itwith PCFG model.	5	K2

QP25EP1_311 | 28-May-2025 9:11:23 AM | 49.42.36.244

QP25EP1_311 | 28-May-2025 9:11:23 AM | 49.42.36.244